Phase 1 | Survey and spatial inputs

Field records

23 forest records
Ngangao (9) and Mbololo (14)
Juveniles, larger stems and cover

Focal species

Coffea fadenii
Afrocarpus usambarensis
Field and GBIF occurrences screened for modelling

Spatial layers

WorldClim 2.1 (2.5 arc-min)
Kenya prediction grid
Protected Planet polygons

Phase 2 | Forest comparisons

RQ1 Woody composition

Record-by-taxon matrices
Richness and family shares
Jaccard and Bray-Curtis
NMDS

RQ2 Vegetation structure

DBH, height, basal area and cover
Growth-form shares and IVI
Repeat IVI for DBH >5 cm

RQ3-4 Focal species

Counts and detection frequency
Lorenz curves and Gini
Size-class overlap and header-stem sensitivity

Phase 3 | Climatic suitability | RQ5

Fit Maxent models

Screened occurrences and climate
75 and 175 km buffers
Unthinned, 8 and 16 km thinning

Evaluate and compare

Latitude-band holdout omission
10th and 5th percentiles
Spearman and area Jaccard

Map and shortlist

Suitability and agreement
Majority vote at the 10th percentile in every scenario
Within training ranges in every fit

Phase 4 | Protected-area coverage | RQ6

Select suitability layers

From RQ5, narrow and broad baselines at the 10th percentile
and the across-scenario shortlist

Intersect suitable cells

Dissolve protected-area polygons
and clip to the Kenya boundary
Weight cells by fraction covered

Report coverage

Overlap area (km²)
Percentage of total suitable area
on the Kenya prediction grid